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United States Patent	12401692
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Sapir; Stav et al.

System and method for quantification of standard compliance and risk tolerance to select remediation

Abstract

Methods and systems for managing computing infrastructure compliance with standards are disclosed. The computing infrastructure may provide computer implemented services that may be at elevated risk if the computing infrastructure fails to comply with various standards such as security or redundancy standards. To manage compliance with standards, a cross-standard compliance coverage model may be used. The cross-standard compliance coverage model may use information regarding infrastructure components of the computing infrastructure to ascertain compliance with any number of standards. The information and risk tolerance may be used to identify a risk profile presented by computing infrastructure.

Inventors:	Sapir; Stav (Beer Sheba, IL), Balin; Maxim (Gan-Yavne, IL)
Applicant:	Dell Products L.P. (Round Rock, TX)
Family ID:	1000008779958
Assignee:	Dell Products L.P. (Round Rock, TX)
Appl. No.:	18/304777
Filed:	April 21, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240356976 A1	Oct. 24, 2024

Publication Classification

Int. Cl.:	H04L9/40 (20220101)
U.S. Cl.:	

Field of Classification Search**USPC:** None**References Cited****U.S. PATENT DOCUMENTS**

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Primary Examiner: Joo; Joshua

Attorney, Agent or Firm: WOMBLE BOND DICKINSON (US) LLP

Background/Summary

FIELD

(1) Embodiments disclosed herein relate generally to security. More particularly, embodiments disclosed herein relate to systems and methods to secure distributed systems.

BACKGROUND

(2) Computing devices may provide computer-implemented services. The computer-implemented services may be used by users of the computing devices and/or devices operably connected to the computing devices. The computer-implemented services may be performed with hardware components such as processors, memory modules, storage devices, and communication devices. The operation of these components and the components of other devices may impact the performance of the computer-implemented services.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments disclosed herein are illustrated by way of example and not limitation in the figures of the accompanying drawings in which like references indicate similar elements.

- (2) FIG. 1 shows a block diagram illustrating a system in accordance with an embodiment.
- (3) FIGS. 2A-2D show data flow diagrams illustrating data flows in accordance with an embodiment.
- (4) FIGS. 3-4 show flowchart diagrams illustrating methods in accordance with an embodiment.
- (5) FIG. 5 shows a data flow diagram in accordance with an embodiment.
- (6) FIG. 6 shows a flowchart diagram illustrating a method in accordance with an embodiment.
- (7) FIG. 7 shows a data flow diagram in accordance with an embodiment.
- (8) FIG. 8A shows a diagram of a rating system in accordance with an embodiment.
- (9) FIG. 8B shows a diagram of a dashboard in accordance with an embodiment.
- (10) FIG. 9 shows a data flow diagram in accordance with an embodiment.
- (11) FIG. 10 shows a block diagram illustrating a data processing system in accordance with an embodiment.

DETAILED DESCRIPTION

- (12) Various embodiments will be described with reference to details discussed below, and the accompanying drawings will illustrate the various embodiments. The following description and drawings are illustrative and are not to be construed as limiting. Numerous specific details are described to provide a thorough understanding of various embodiments. However, in certain instances, well-known or conventional details are not described in order to provide a concise discussion of embodiments disclosed herein.
- (13) Reference in the specification to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in conjunction with the embodiment can be included in at least one embodiment. The appearances of the phrases “in one embodiment” and “an embodiment” in various places in the specification do not necessarily all refer to the same embodiment.
- (14) References to an “operable connection” or “operably connected” means that a particular device is able to communicate with one or more other devices. The devices themselves may be directly connected to one another or may be indirectly connected to one another through any number of intermediary devices, such as in a network topology.
- (15) In general, embodiments disclosed herein relate to methods and systems for managing computing infrastructure. The computing infrastructure may provide computer implemented services. Providing the computer implemented services may present risk of undesired disclosure of data and/or other undesired outcomes.
- (16) To mitigate the risks, various standards for infrastructure components of the computing infrastructure may be enforced. The standards may specify various requirements for the infrastructure components.
- (17) To manage compliance with the standards, a cross-standard compliance coverage model may be used. The cross-standard compliance coverage model may use information regarding the infrastructure components to ascertain whether the computing infrastructure meets the standards.
- (18) To obtain the information used by the cross-standard compliance coverage model, agents may be deployed and configured based on mappings included in the cross-standard compliance coverage model. The agents may monitor the infrastructure components and generate compliance information elements.
- (19) The compliance information elements may include information usable to ascertain whether the infrastructure component complies with various portions of different standards (e.g., security standards, redundancy standards, etc.). The information may be used to update the cross-standard compliance coverage model in near real time. Consequently, changes in compliance with any number of standards may be identified for computing infrastructure.
- (20) The changes in compliance may be identified at both a granular and macro level. For example, quantifications regarding compliance with different parameters of requirements of standards may be obtained. These quantifications may be used to obtain other quantifications for the standard as a

whole (e.g., using averaging, weighted averages, etc.). The granular quantifications may also be used to identify actions that when performed may be likely to improve compliance with a standard. For example, parameters for which compliance is failed may be used as a basis for an action. Any number of actions may be identified for different parameters of various requirements of standards. (21) When changes in compliance are identified, various actions may be performed to manage the impact of the changes in compliance. The actions performed may include some or all of the identified actions. The selection process for the actions may be based on criteria that may be based on the level of compliance with the standards (e.g., lower compliance may cause more actions of the identified actions to be selected for performance, e.g., to meet goals with respect to levels of compliance with standards or portions thereof). By doing so, embodiments disclosed herein may provide computing infrastructure that is less likely to suffer from risks in providing computer implemented services.

(22) To customize responses to compliance failure, the actions may be selected based on confidentiality-integrity-availability (CIA) classifications for different infrastructure components. The CIA classifications may allow an operator to specify the level of concern with respect to confidentiality, integrity, and availability provided by the respective infrastructure component. The operator (or an automated process) may ascribe these levels of concern based on the function of the infrastructure component, data which will be accessed by the infrastructure component, and/or other factors. In this manner, the compliance failures and CIA ratings may allow an operator of infrastructure to better understand the risk profile of their infrastructure, and initiate appropriate remedial activity to align the level of compliance with standards to these risks.

(23) In an embodiment, a method for managing computing infrastructure is disclosed. The method may include obtaining a compliance information element for an infrastructure component of the computing infrastructure; dynamically processing the compliance information element to update a cross-standard compliance coverage model to obtain an updated cross-standard compliance coverage model; obtaining, using the updated cross-standard compliance coverage model, standard compliance data for a security standard enforced on the infrastructure; making a determination, based on the standard compliance data, a confidentiality-integrity-availability classifications for the infrastructure, and a rating system, whether the infrastructure has undergone a change in compliance with the security standard; in an instance of the determination where the infrastructure has undergone a change in compliance with the security standard: performing an action set to manage an impact of the change in compliance with the security standard.

(24) Making the determination may include obtaining a likelihood estimate for the infrastructure component using the standard compliance data; obtaining, using the likelihood estimate, the confidentiality-integrity-availability classifications, and the rating system, a confidentiality-integrity-availability rating for the infrastructure component; and using the confidentiality-integrity-availability rating to make the determination.

(25) Using the confidentiality-integrity-availability rating to make the determination may include presenting, using the confidentiality-integrity-availability rating, a dashboard; obtaining, using the dashboard, user input responsive to the confidentiality-integrity-availability rating; and using the user input to make the determination.

(26) The confidentiality-integrity-availability classifications may include a first classification for confidentiality, the first classification indicating a numerical value within a range for confidentiality concern for the infrastructure component.

(27) Obtaining the confidentiality-integrity-availability rating may include obtaining a quantification using the numerical value and the likelihood estimate.

(28) Obtaining the confidentiality-integrity-availability rating may also include obtaining a second quantification using a second numerical value and the likelihood estimate, the second numerical value being within a range for integrity concern for the infrastructure component.

(29) The confidentiality-integrity-availability rating may include three sub-ratings, the sub-ratings

being for confidentiality concern, integrity concern, and availability concern for the infrastructure component, respectively.

(30) Each of the three sub-ratings may be set by a manager of the infrastructure component.

(31) In an embodiment, a non-transitory computer readable media is provided. The non-transitory computer readable media may include instructions that when executed by a processor cause the computer-implemented method to be performed.

(32) In an embodiment, a data processing system is provided. The data processing system may include the non-transitory media and a processor, and may initiate performance of the method when the computer instructions are executed by the processor.

(33) Turning to FIG. 1, a block diagram illustrating a system in accordance with an embodiment is shown. The system shown in FIG. 1 may provide computer-implemented services. The computer implemented services may include any type and quantity of computer implemented services. For example, the computer implemented services may include data storage services, data acquisition services, communication services, and/or any other type of service that may be implemented with a computing device.

(34) To provide the computer implemented services, the system of FIG. 1 may include computing infrastructure **100**. Computing infrastructure **100** may include any number of infrastructure components **102-104** that may provide the computer implemented services. The infrastructure components may cooperatively and/or independently provide all, or portions, of the computer implemented services.

(35) While providing the computer implemented services, infrastructure components **102-104** may (i) be subject to interference (e.g., attacks) by malicious entities, (ii) be subject to errors in processing or other issues, etc. The interference, errors in processing, and/or other issues arising in the computer implemented services may cause the computer implemented services to diverge from nominal performance. For example, the interference by malicious entities may cause data that would otherwise remain private to be disclosed to entities that are not authorized to access the data. Similarly, errors in processing may cause some data that may otherwise be expected to be accessible in the future to become inaccessible (e.g., due to corruption, etc.).

(36) To manage impacts of interference, errors in processing, and/or other issues, various standards for operation of computing infrastructure **100** may be established. The standards may relate to, for example, security, reliability, regulatory requirements, and/or other aspects regarding the operation and/or expectations for operation of infrastructure components **102-104**.

(37) However, the standards for operation of computing infrastructure **100** may not explicitly specify how the infrastructure components are to operate. Rather, the standards may set expectations or goals, but may also not specify how to identify whether the standards or expectations (e.g., requirements which may be multidimensional and include various parameters) are met. Further, different standards may specify the standards and/or expectations in different manners.

(38) Additionally, failure to comply with any of the standards for any duration of time may not be acceptable. For example, the standards may specify that the standards and expectations are to be continually met.

(39) In general, embodiments disclosed herein may provide methods, systems, and/or devices for providing computer implemented services that are more likely to be provided in manners that comply with any number of standards. The disclosed systems may provide the computer implemented services using a standards management framework. The standards management framework may measure compliance of infrastructure components used to provide the computer implemented services with corresponding standards. If the operation of the infrastructure components diverges from the standards, then the system of FIG. 1 may take action to manage impact of the divergence of the operation of the infrastructure components from the standard.

(40) Divergences from the standards may be characterized granularly (e.g., on a per requirement of

a standard level) and at a macro level (e.g., on a per standard level). For example, quantifications (e.g., percentage levels) regarding the level of divergence may be obtained. Additionally, the divergences at the granular level may be used to identify remediation actions that when performed are likely to reduce the divergences. In this manner, both granular and macro level views of compliance, as well as options for improving compliance, may be obtained.

(41) To ascertain whether and to what extent remediations for infrastructure will be performed, confidentiality-integrity-availability classifications for the infrastructure may be taken into account, the classification may be part of an infrastructure management plan and may ascribe different levels for different concerns to the infrastructure. The classifications may be set based on a range, and different portions of the infrastructure may be rated differently to allow for individualized remediation behavior to be implemented for different type of compliance failures.

(42) By doing so, a system in accordance with embodiments disclosed herein may be more likely to provide computer implemented services in a manner that meets any number of standards. The system may do so by continuously monitoring for compliance with the standards. By continuously monitoring, the resulting computer implemented services may be more likely to be nominal in performance. Thus, embodiments disclosed herein may address, among others, the technical problem of security in distributed systems. The disclosed embodiments may address security in distributed systems through dynamic analysis of compliance with any number of standards for infrastructure components that provide computer implemented services.

(43) To provide the above noted functionality, the system of FIG. 1 may include computing infrastructure **100** and infrastructure management system **110**. Each of these components is discussed below.

(44) Computing infrastructure **100** may provide the computer implemented services. To do so, computing infrastructure **100** may include any number of infrastructure components **102-104**. The infrastructure components may include objects that contribute to performance of the computer implemented services and that may be monitorable. For example, any of the infrastructure components may include hosted entities (e.g., executing programs, data structures, etc.) and/or host devices (e.g., hardware such as data processing systems) that contribute to performance of the computer implemented services.

(45) The operation of the infrastructure components may be subject to any number of standards (e.g., that attempt to retain performance of computer implemented services in a nominal manner). To identify whether the operation of the infrastructure components indicates compliance or non-compliance with any of the standards, agents may monitor the infrastructure components. Through the monitoring, information (e.g., facts) relevant to whether standards are complied with or violated by the computing infrastructure may be obtained. Refer to FIGS. 2B-2D for additional details regarding obtaining of relevant information for infrastructure components.

(46) Infrastructure management system **110** may monitor computing infrastructure **100** for compliance with any number of standards, and may take action to manage impacts of non-compliance of computing infrastructure **100**. By doing so, the computer implemented services provided by computing infrastructure **100** may be more likely to be provided in a nominal manner.

(47) To monitor computing infrastructure **100** for compliance, infrastructure management system **110** may (i) deploy and use agents to collect information from infrastructure components **102-104**, (ii) use the collected information to update a cross-standard compliance coverage model, (iii) use the updated cross-standard compliance coverage model to ascertain whether computing infrastructure **100** is in or out of compliance with any number of standards, and/or (iv) initiate performance of any number of actions to manage impacts of compliance failures by computing infrastructure **100**. Refer to FIG. 2C for additional details regarding deploying agents. Refer to FIGS. 2A and 2D for additional details regarding updating of cross-standard compliance coverage models and identifying standard compliance. Refer to FIG. 5 for additional details regarding identifying levels of compliance with various standards.

(48) The specific remediation actions may be selected based on confidentiality-integrity-availability (CIA) classifications for infrastructure. The CIA classifications may establish levels of concern for the infrastructure and may cause different remediation actions to be performed for similar compliance failures. The CIA classifications and identified compliance failures may be used to select the remediation actions to perform. Refer to FIGS. 7-8B for additional details regarding use of CIA classifications for managing infrastructure.

(49) When providing their functionality, any of computing infrastructure **100** and infrastructure management system **110** may perform all, or a portion, of the methods illustrated in FIGS. 3-4, 7, and 9.

(50) Any of computing infrastructure **100** and infrastructure management system **110** may be implemented using a computing device (also referred to as a data processing system) such as a host or a server, a personal computer (e.g., desktops, laptops, and tablets), a “thin” client, a personal digital assistant (PDA), a Web enabled appliance, a mobile phone (e.g., Smartphone), an embedded system, local controllers, an edge node, and/or any other type of data processing device or system. For additional details regarding computing devices, refer to FIG. 10.

(51) Any of the components illustrated in FIG. 1 may be operably connected to each other (and/or components not illustrated) with communication system **120**. In an embodiment, communication system **120** includes one or more networks that facilitate communication between any number of components. The networks may include wired networks and/or wireless networks (e.g., and/or the Internet). The networks may operate in accordance with any number and types of communication protocols (e.g., such as the internet protocol).

(52) While illustrated in FIG. 1 as including a limited number of specific components, a system in accordance with an embodiment may include fewer, additional, and/or different components than those illustrated therein. For example, a system may include any number of computing infrastructure **100** and/or other components not shown in FIG. 1. Any of the instances may perform similar and/or different functions performed by other instances.

(53) Turning to FIG. 2A, a first data flow diagram in accordance with an embodiment is shown. In the data flow diagram, flows of data and processing of data are illustrated. Processes (e.g., executing software) are illustrated using a first set of elements (e.g., **220**, **222**) and data structures (e.g., tables, linked lists, databases, etc.) are illustrated using a second set of elements (e.g., **200**-**214**). Similar elements are used in FIGS. 2B-2C to distinguish processes and data structures in data flow diagrams.

(54) To ascertain compliance of computing infrastructure with any number of standards, infrastructure information **200** may be obtained. Infrastructure may include any number of compliance information elements (e.g., **202**-**204**). Compliance information elements **202**-**204** may include information regarding an infrastructure component. The information may be mapped by cross-standard compliance coverage model **206** to portions of any number of standards (e.g., security standards, operational standards, regulatory standards, etc.). The mappings may also indicate how the content of each compliance information element shows either compliance or non-compliance with the corresponding portions of the standards.

(55) Infrastructure information **200** may be obtained by receiving it from agents that monitor the infrastructure components. Refer to FIG. 2B for additional details regarding compliance information elements **202**-**204**.

(56) Infrastructure information **200** may be used in compliance validation process **220** to (i) update a state of cross-standard compliance coverage model **206** and (ii) establish compliance status **208**. For example, compliance validation process **220** may use the mappings of cross-standard compliance coverage model to process compliance information elements **202**-**204**. The processing may establish whether each compliance information element indicates whether the computing infrastructure complies with corresponding portions of standards. The state of cross-standard compliance coverage model **206** may be updated based on the determined compliance of the

computing infrastructure. Thus, the state of cross-standard compliance coverage model **206** may indicate compliance and non-compliance with any number of portions of any number of standards. Refer to FIG. **2D** for additional details regarding the state of cross-standard compliance coverage model **206**.

(57) The state of cross-standard compliance coverage model, once updated, may be used to generate compliance status **208** as part of compliance validation process **220**. Compliance status **208** may reflect compliance of the computing infrastructure with respect to any number of standards.

(58) Compliance status may include any amount of standard compliance data **210-212**. Each portion of standard compliance data may indicate compliance with portions of a standard. For example, standard compliance data **210** may be a data structure that specifies which portions of a standard that computing infrastructure is in compliance with and other portions of the standard that computing infrastructure is out of compliance with.

(59) For example, one of the standards may be the National Institutes of Standards and Technology 800-53 standard, which sets expectations regarding computing infrastructure for security purposes. A corresponding standard compliance data (e.g., **210**) for the 800-53 standard may indicate which portions of the standard the computing infrastructure is in compliance and other portions of the standard that the computing infrastructure is out of compliance. Different standard compliance data may correspond to different standards.

(60) The standards compliance data may indicate (e.g., quantify) compliance at both granular and/or macro levels. Refer to FIG. **5** for additional details regarding quantifying compliance.

(61) Once generated, compliance status **208** may be used in compliance management process **222** to manage impacts of compliance failures. For example, during compliance management process **222**, various actions keyed to compliance failures specified by compliance status **208** may be performed to attempt to manage the impact of these compliance failures. The actions may include, for example, sending various notifications, modifying operation of infrastructure components of the computing infrastructure, disabling various infrastructure components, etc. Refer to FIG. **5** for additional details regarding identifying actions that when performed may improve compliance with standards. Refer to FIG. **7** for additional details regarding selecting which actions of the identified actions to perform during remediation.

(62) To implement the actions, compliance management process **222** may generate management actions **214**. Management actions **214** may specify actions (e.g., to manage the impact) to be performed by various entities. Management actions **214** may be distributed to corresponding entities that will perform the actions. Once distributed, the actions may be performed by the entities.

(63) For example, as will be discussed further with respect to FIG. **5**, actions that may improve compliance may be identified initially based on granular quantification of compliance with different requirements of standards. During and/or along with quantification, actions that may improve compliance with each requirement may be identified. Management actions **214** may be identified based on the actions that are identified as likely improving compliance with the corresponding requirement. Then performance of actions may be prioritized based on CIA classifications and/or rating systems. Refer to FIG. **7** for additional details regarding prioritizing actions, and FIG. **8A** for additional details regarding rating systems.

(64) Turning to FIG. **2B**, a second data flow diagram in accordance with an embodiment is shown. In the data flow diagram, flows of data and processing of data are illustrated. To acquire information usable to ascertain whether computing infrastructure is in compliance with standards, an agent **230** may be utilized.

(65) For example, infrastructure component **102** may be hosted by data processing system **290**. Other entities **103** may also be hosted by data processing system **290**. The other entities may include, for example, operating systems, drivers, applications, etc.

(66) Agent **230** may perform monitoring process **232** with respect to infrastructure component **102** and other entities **103**. During monitoring process **232** information regarding operation of infrastructure component **102** and other entities **103** may be collected. Through monitoring process **232**, configuration **240**, state **242**, and audit result **244** may be obtained.

(67) Configuration **240** may include information regarding the configuration of infrastructure component **102** and/or other entities **103**. The configuration may reflect various operations that may be selected and which influence operation of infrastructure component **102** and other entities **103** (e.g., such as requirements for use of secure connections, encryption, etc.). This information may be obtained by, for example, reading it from files that define configurations, requesting it from infrastructure component **102** and/or other entities **103**, and/or via other methods.

(68) State **242** may include information regarding the state of infrastructure component **102** and/or other entities **103**. The state may reflect the operating state of infrastructure component **102** and other entities **103**, and/or the state of the hardware support operation of infrastructure component **102** and/or other entities. This information may be obtained by, for example, requesting from management entities such as operating systems, requesting it from hardware components, reading various registers and/or files, and/or via other data collection methods.

(69) Audit result **244** may include information regarding an outcome of an audit process performed on data processing system **290** and/or hosted entities. The audit process may be performed by agent **230** and/or other entities (e.g., another application may perform it at the request of agent **230**). The audit result may reflect whether and to what extent data processing system and/or hosted entities passed the audit, or portions thereof. This information may be obtained by, for example, receiving it from another entity, through performance of the audit process, and/or via other data collection methods.

(70) The information obtained through monitoring process **232** may be used to obtain compliance information element **236**. For example, compliance information generation process **234** may ingest the information and generate compliance information element **236**. Compliance information element **236** may be a data structure having a standardized format that allows for relevant mappings maintained by the cross-standard compliance coverage model to be identified and used to update the state of the cross-standard compliance coverage model.

(71) The specific quantities monitored by monitoring process **232** and the information included by compliance information element **236** may be configured. Refer to FIGS. 2C-2D for additional details regarding configuration for compliance information element generation.

(72) Once obtained, compliance information element **236** may be provided to infrastructure management system **110** for updating of the cross-standard compliance coverage model.

(73) Turning to FIG. 2C, a third data flow diagram in accordance with an embodiment is shown. In the data flow diagram, flows of data and processing of data are illustrated. To acquire information usable to update the state of cross-standard compliance coverage model, an agent may be configured using configuration data **252**.

(74) Configuration data **252** may specify the information to be monitored by an agent, and how the information is to be used to generate compliance information elements. Configuration data **252** may be generated through configuration data generation process **250**.

(75) Configuration data generation process **250** may be performed when a new infrastructure component is identified for compliance monitoring. When new infrastructure components are identified, configuration data generation process **250** may be performed. During configuration data generation process **250**, mappings from cross-standard compliance convergence model that specify the information which needs to be included in compliance information elements from the infrastructure component may be identified. Different mappings may be tagged with respect to different infrastructure components such that relevant mappings may be identified based on the infrastructure components. The identified mappings may be used to identify the content and format of the compliance information elements.

(76) A procedure library or other data structure which defines monitoring procedures based on information to be included in compliance information elements may be used to identify monitoring procedures to be implemented by corresponding agents.

(77) Once the procedures and informational content for compliance information elements, then configuration data **252** may be generated. After generation, configuration data **252** may be used to configure a corresponding agent. For example, configuration data **252** may be sent to the agent. The agent may configure its operation based on configuration data **252**.

(78) Turning to FIG. **2D**, a fourth data flow diagram in accordance with an embodiment is shown. In the data flow diagram, flows of data and processing of data are illustrated. Additionally, various standards **260-264** of the cross-standard compliance coverage model are illustrated using a series of circles. Each circle corresponds to a portion of the respective standard (which may specify any number of requirements (e.g., **270**) for compliance with the portion of the standard, each circle may represent a requirement, and each requirement may include parameters that define various sub-requirements of the requirement), and each circle is associated with a portion of the state of the cross-standard compliance coverage model. While illustrated with a finite number of circles corresponding to portions of standards, it will be appreciated that a standard may include any number of portions.

(79) Additionally, in FIG. **2D**, lines terminating in arrows represent mappings (e.g., **272**) that define relationships between information and portions of the state of cross-standard compliance coverage model. For example, each mapping may specify (i) information on which a portion of the state is based, (ii) rules for identify the portion of the state based on the information, (iii) identities of infrastructure components to which the mapping is applicable, and/or other information usable to identify relevant information and identify portions of the state of the cross-standard compliance coverage model.

(80) For example, in a portion of a standard related to networking, a mapping may specify that (i) information regarding ports is relevant, (ii) that the ports must be inactive to comply with the portion of the standard, and (iii) the mapping is relevant to network components of computing infrastructure. In this example, configuration information for an agent may be generated that specifies collection of the information regarding the ports, and inclusion of the information in a compliance information element (e.g., **236**).

(81) When received by infrastructure management system **110**, the information element may be processed based on the rules specified in the mapping. In this example, the rules may specify that the computing infrastructure is in or out of compliance depending on whether the port is open. For purposes of example, if mapping **272** specifies that the port is to be closed, but compliance information element **236** specifies that the port is open, then the portion of the state associated with standard **260** may be modified to reflect that the computing infrastructure is out of compliance.

(82) Other mappings of the cross-standard compliance coverage model may specify that some portions of standards depend on compliance with portions of other standards. For example, in FIG. **2D**, some arrows representing mappings extend from the representation of the portion of standard **260** to the representations of the portions of standard **262**. These two standards may be different, but may define compliance with portions of standard **262** in terms of compliance with portions of standard **260**. In this manner, the cross-standard compliance coverage model may propagate changes in compliance across multiple standards without needing to obtain other compliance information elements and/or perform other type of computations to evaluate compliance information elements. Accordingly, when standard compliance data (e.g., **210**) is generated for a standard, the information maintained by the cross-standard compliance coverage model regarding the standard may be kept up to date and propagated to identify the states compliance with any number of standards.

(83) The mappings of the cross-standard compliance coverage model may be established by, for example, subject matter experts. As standards are added or removed from the cross-standard

compliance coverage model, the mappings may be correspondingly updated.

(84) As discussed above, the components of FIG. 1 may perform various methods to manage the compliance with various standards to improve the likelihood of desirable computer implemented services being provided. FIGS. 3-4 and 6 illustrate methods that may be performed by the components of the system of FIG. 1. In the diagrams discussed below and shown in FIGS. 3-4 and 6, any of the operations may be repeated, performed in different orders, and/or performed in parallel with or in a partially overlapping in time manner with other operations.

(85) Turning to FIG. 3, a flow diagram illustrating a method for managing the impact of changes in compliance with standards in accordance with an embodiment is shown. The method may be performed in part by computing infrastructure 100, infrastructure management system 110, and/or other components of the system shown in FIG. 1.

(86) At operation 300, a compliance information element for an infrastructure component is obtained. The compliance information element may be obtained from an agent. The agent may generate the compliance information element. The compliance information element may be obtained by (i) receiving it via a message, (ii) reading it from storage, and/or via other methods.

(87) At operation 302, the compliance information element is dynamically processed to update a cross-standard compliance coverage model. The compliance information element may be dynamically processed by (i) identifying mappings associated with content of the compliance information element, (ii) identifying processing rules specified by the mappings, and (iii) processing the content of the compliance information elements based on the processing rules to update the state (e.g., compliance state indicating whether computing infrastructure complies with any number of standards) of the cross-standard compliance coverage model. The processing may update the state of compliance with any number of portions of standards of the cross-standard compliance coverage model. For example, as discussed above, changes in compliance with portions of a first standards may be used to automatically identify changes in portions of other standards (e.g., based on inter-standard mappings rather than compliance information element to standard mappings).

(88) At operation 304, compliance standard data for a security standard is obtained using the updated cross-standard compliance coverage model. The compliance standard data may be obtained by (i) instantiating a data structure and (ii) populating the data structure with information reflecting compliance with the security standard based on corresponding portions of the cross-standard compliance coverage model. The resulting data structure may specify (i) portions of the security standard for which the security infrastructure is in compliance and/or (ii) other portions of the security standard for which the security infrastructure is out of compliance.

(89) The compliance standard data may be obtained via the method illustrated in FIG. 5, and/or other manners.

(90) At operation 306, a determination is made regarding whether a change in compliance of the computing infrastructure has occurred. The determination may be made based on the compliance standard data (and/or compliance standard data for any number of standards). For example, if the compliance standard data indicates that the computing infrastructure has fallen out of compliance with one or more portions of one or more standard (and/or other standards), then it may be determined that a compliance change has occurred.

(91) If a compliance change has occurred, then the method may proceed to operation 308. Otherwise the method may end following operation 306.

(92) At operation 308, an action set may be performed to manage an impact of change in compliance of the computing infrastructure. The action set may be performed by identifying actions to be performed based on one or more compliance failures specified by the compliance standard data. The actions may include, for example, sending notification or other types of information regarding the compliance failures to one or more recipients (e.g., administrators, decision makers, etc.), documenting the compliance failures, initiating automatic attempts to

remediate the compliance failures, initiating shut down/quarantining/other actions to limit exposure of the portions of the computing infrastructure that has failed compliance with the standards, and/or other actions that may reduce the impact of the compliance changes. The actions may be identified by, for example, performing a lookup in a data structure that specifies actions to be performed for different types of compliance failures/changes in compliance. Performance of the actions may then be initiated (e.g., by generating and sending instructions regarding the actions to entities tasked with performing the actions).

(93) Once identified, some of the actions may be selected for performance based on CIA classifications and/or rating systems for compliance. For example, the CIA classifications may classify infrastructure for different concerns, and the level of the concerns may be used to determine which actions to perform. The actions may be selected using the method illustrated in FIG. 9.

(94) The method may end following operation **308**.

(95) Turning to FIG. 4, a flow diagram illustrating a method for obtaining compliance information elements in accordance with an embodiment is shown. The method may be performed in part by computing infrastructure **100**, infrastructure management system **110**, and/or other components of the system shown in FIG. 1.

(96) At operation **400**, an infrastructure component may be identified for standard monitoring. The infrastructure component may be identified by (i) receiving information regarding the infrastructure component from another entity, (ii) by reading the information from storage, (iii) by obtaining user input indicating the infrastructure components, and/or via other methods.

(97) At operation **402**, mappings of a cross-standard compliance coverage model related to the infrastructure components are identified. The mappings may be identified based on a type of the infrastructure component. Mappings of the cross-standard compliance coverage model may be associated with different types of infrastructure components. The mappings may be identified based on the associations.

(98) At operation **404**, a configuration for an agent for the infrastructure component is generated based on the identified mappings. The configuration for the agent may be generated by (i) instantiating a data structure, (ii) populating the data structure with information regarding quantities used by the mappings to identify compliance with standards of the cross-standard compliance coverage model, and/or (iii) populating the data structure with information regarding a format for compliance information elements.

(99) At operation **406**, operation of an instance of the agent associated with the infrastructure component is updated based on the configuration. The operation of the instance of the agent may be updated by (i) providing the instance of the agent with the data structure (e.g., the agent may self-configure), (ii) providing a control plane or other management entity with the data structure (e.g., which may configure the operation of the agent using the data structure), (iii) by instantiating the agent with the data structure (e.g., by sending instructions to a data processing system that hosts the infrastructure component), and/or via other methods.

(100) The method may end following operation **406**.

(101) Once configured, the agent may collect information as specified by the configuration, may generate compliance information elements as specified by the configuration, and may forward or otherwise provide the compliance information elements to other entities for use.

(102) Using the methods illustrated in FIGS. 3-4, embodiments disclosed here may facilitate near real time compliance analysis for computing infrastructure. The compliance analysis may then be used to manage the impact of changes in compliance. Consequently, computer implemented services by the compliance infrastructure may be more likely to meet expectations.

(103) Turning to FIG. 5, a fifth data flow diagram in accordance with an embodiment is shown. In the data flow diagram, flows of data and processing of data are illustrated. Processes (e.g., executing software) are illustrated using a first set of elements (e.g., **504**, **510**, **512**) and data

structures (e.g., tables, linked lists, databases, etc.) are illustrated using a second set of elements (e.g., **500**, **502**, **506**, **508**, **514**, **516**).

(104) To obtain compliance data for a standard, requirements of the standard may be evaluated on a parameter by parameter basis. In FIG. 5, an evaluation for a requirement **502** of a standard is shown. Requirement **502** may include any number of parameters. Each parameter may specify, for example, a characteristic of infrastructure that must be met for the requirement to be met.

(105) To evaluate to an extent that requirement **502** is met, parameter compliance analysis process **504** may be performed. During parameter compliance analysis process **504**, the parameters of requirements **502** may be individually evaluated based on either an acquired compliance information element or propagated information (e.g., **500**) from another standard. As discussed with respect to FIG. 2D, some requirements of standards may be based on the state of compliance with other standards.

(106) Passed parameters **506** and failed parameters **508** may be identified via parameter compliance analysis process **504**. The number of failed parameters **508** and passed parameters **506** may be used in quantification process **510**. During quantification process **510**, a quantification for the extent of compliance with requirement **502** may be identified. For example, a ratio of the passed parameters to the total parameters of requirement **502** may be obtained and added to requirement compliance report **514**.

(107) Requirement compliance report **514** may quantify the extent to which requirement **502** was met. While not shown, requirement compliance report **514** may be aggregated with other requirement compliance reports to obtain a report reflecting the extent of compliance with any number of requirements of a standard. Additionally, a quantification based on the level of compliance with each of the requirements may be obtained. For example, the quantification may be obtained by calculating an average, weighted average, and/or other quantity based on the extent of compliance with the requirements of the standard. The extent of compliance for each requirement and the standard as a whole may be standard compliance data (e.g., **210-212**) for the standard.

(108) Failed parameters **508** may also be used in remediation analysis process **512** to identify remediation actions **516**. Remediation actions **516** may be identified by performing lookups or other types or processing based on failed parameters **508**. The processing may return actions which when performed by infrastructure may be likely to cause the parameter to be passed. For example, the actions may modify the manner in which the infrastructure operates. The change in operation of the infrastructure may be likely to cause the parameter to be passed during a future parameter compliance analysis process. Remediation actions **516** may include any number of actions associated with any number of failed parameters **508**.

(109) Once obtained, the remediation actions may be screened to identify a final set of actions to perform as a remediation action set. Refer to FIG. 7 for additional details regarding identifying a final remediation action set.

(110) Turning to FIG. 6, a flow diagram illustrating a method for obtaining compliance standard data in accordance with an embodiment is shown. The method shown in FIG. 6 may be an expansion of operation **304** shown in FIG. 3. The method may be performed in part by computing infrastructure **100**, infrastructure management system **110**, and/or other components of the system shown in FIG. 1.

(111) At operation **600**, a requirement of a first security standard is obtained. The requirement may be obtained by reading the requirement from storage or receiving the requirement from another device.

(112) At operation **602**, parameters of the requirement are identified. The parameters may be identified by parsing the requirement.

(113) At operation **604**, a change in compliance with at least one of the parameters is identified based on a compliance information element or information propagated from a requirement of a second security standard. The change in compliance may be identified using the compliance

information element when the cross-standard compliance coverage model specifies the requirement in terms of the content of the compliance information element, and the change in compliance may be identified using information propagated from the requirement of the second security standard when the cross-standard compliance coverage model specifies the requirement in terms of the content of the compliance information element. As discussed with respect to FIG. 2D, the requirement may be specified in terms of either of these data structures to facilitate identification of compliance with any number of standards.

(114) The change in compliance may be identified by performing processing as specified by mappings of the cross-standard compliance coverage model. The result of the processing may indicate compliance with respect to each of the at least one of the parameters. The change in compliance may be identified by comparing the new state of the parameters to the old states (e.g., a change in state may indicate a change in compliance).

(115) At operation **606**, a first quantification of compliance with the requirement is updated based on the change in the compliance with the at least one of the parameters. The quantification may be updated by dividing the number of parameters which are complied with based on the total number of parameters of the requirement. It will be appreciated that some parameters may be weighted, and/or other modifiers may be present depending on a desired goal for the quantification.

(116) At operation **608**, an action that when performed increases the first quantification is identified based on the change in the compliance with the at least one of the parameters. The action may be identified by performing a lookup or other processing operation to identify the action. The processing may use the at least one of the parameters, or a portion thereof, as a key. For example, an identifier of one of the at least one of the parameters may be used to perform a lookup, which may return the action.

(117) At operation **610**, a quantification of compliance with the first security standard may be updated based on the first quantification. The quantification may be similarly to first quantification. For example, the first quantification may be used to calculate a ratio of total passed parameters divided by the total number of parameters of the first security standard. Like the first quantification, the quantification of the compliance with the first security standard may weight certain parameters to meet various goals.

(118) The method may end following operation **310**.

(119) Thus, using the method illustrated in FIG. 6, both granular and macro level quantification of compliance may be obtained. These quantifications may be used to select how action sets are performed to manage the impact of changes in compliance.

(120) For example, the actions identified in operation **608** may be one of the actions included in an action set. Similarly, the granular and global quantifications may be used to select actions of the action set, may be used as part of the actions in the action set, and/or may otherwise be used to manage the impacts of changes in compliance.

(121) When deciding on the actions to include in the action set, CIA classifications for infrastructure may be taken into account. As noted above, the CIA classifications may specify levels of concern for the infrastructure. These levels of concern may be used, in part, to select the extent of the remediation to be performed.

(122) Turning to FIG. 7, a sixth data flow diagram in accordance with an embodiment is shown. In the data flow diagram, flows of data and processing of data are illustrated. Processes (e.g., executing software) are illustrated using a first set of elements (e.g., **704**) and data structures (e.g., tables, linked lists, databases, etc.) are illustrated using a second set of elements (e.g., **700**, **514**, **702**, **706**).

(123) Different infrastructure components may perform different functions as part of a distributed system. Consequently, compromise of different infrastructure components may have different ramifications regarding confidentiality (e.g., of data), integrity, and availability of functionality provided by the distributed system. Accordingly, it may be more important to ensure a greater level

of compliance with standards for some infrastructure components than for other infrastructure components.

(124) To manage different infrastructure components, CIA classifications **700** for infrastructure components may be maintained. CIA classifications **700** may include classifications for different infrastructure components regarding confidentiality, integrity, and availability. Thus, sub-classifications for each of these separate concerns for each infrastructure component may be included in CIA classifications **700**. The ratings may be established by an operator of the infrastructure, or other interested party.

(125) CIA classifications **700** may be used in compliance analysis processes **704** through which ratings **706** are obtained. During compliance analysis process **704**, ratings regarding confidentiality, integrity, and availability of an infrastructure component are obtained. These ratings may be quantitative (e.g., on a scale such as 1-10) or qualitative (e.g., poor, average, good). The ratings may be used to select the extent of remediations to be performed, as well as ascertain whether a compliance change for the infrastructure component has occurred.

(126) During compliance analysis process **704**, requirement compliance reports (e.g., **514**) for an infrastructure component may be ingested and used to calculate a likelihood of an compromise occurring. As discussed above, a requirement compliance report may specify granular and macro level quantifications of compliance with a standard. The likelihood may be calculated by subtracting the macro level quantification of compliance with the standard from 100%. For example, if an infrastructure component is rated as 82% compliant in requirement compliance report **514**, then a likelihood of compromise may be calculated as 18% (e.g., $100 - 82 = 18$).

(127) Once obtained, the CIA classifications and likelihood may be used to obtain ratings **706** using rating system **702**. Rating system **702** may specify a formula or other function through which the likelihood and sub-classifications from the CIA classifications may be used to obtain a sub-rating of ratings **706**. Refer to FIG. **8A** for additional details regarding an example rating system.

(128) For example, the CIA ratings may include three numerical values indicating classifications for confidentiality, integrity, and availability of an infrastructure component. Sub-ratings may be calculated using each of these numerical values by separately ingesting the numerical rating and the likelihood into a function. The function may then output a numerical value for the sub-rating. The sub-ratings may be aggregated to obtain ratings **706**. The ratings may reflect the relative level of concern for the risk presented by the infrastructure component. These ratings may be used, for example, (i) to select the extent of remediations that will be performed, (ii) to inform persons of the risks presented by the infrastructure component, and/or for other purposes. Refer to FIG. **8B** for additional details regarding information persons of risks presented by the infrastructure.

(129) Ratings **706** may provide separate numerical scores reflecting the level of risk for confidentiality, integrity, and availability presented by the infrastructure component.

(130) Turning to FIG. **8A**, a diagram of an example rating system in accordance with an embodiment is shown. The example rating system may be usable to rate risk based on CIA classifications and likelihood of compromise. In FIG. **8A**, the rating system may be a binned function of the product of the likelihood and the CIA classification.

(131) For example, the rating system may ascribe different levels of risk based on (i) ranges of likelihood of an infrastructure component being compromised, and (ii) CIA classification values.

(132) The likelihood (e.g., **812**) may be binned into buckets of approximate 20% each (i.e., 0-20%, 21-40%, etc.), and the CIA classification (e.g., **814**) may be binned into different ranges A-E. Thus, if the CIA classifications includes values that fall within a range of 1-5, a value of 0-1 may fall into class A, a value greater than 1 but less than 2 may fall into class B, etc. The rating system may ascribe a different rating (e.g., **810**) based on the product of the classification range and likelihood range. In FIG. **8A**, different fill patterns are used to represent the different ratings ascribed to different combinations of likelihood range and classification range. As seen in FIG. **8A**, the rating system may ascribe different or similar ratings for different classification ranges and likelihood

ranges. This example rating system may balance level of concern for CAI and likelihood. However, it will be appreciated that a rating system may more heavily weight CIA concern or likelihood, and may include more granularized ratings.

(133) For example, if a CIA classification includes a confidentiality value of 1 for an infrastructure component, and the likelihood of compromise was found to be 19%, then the rating system may ascribe a “low” confidentiality risk rating to the infrastructure component. In contrast, if the CIA classification includes a confidentiality value of 4 for an infrastructure component (e.g., class D), and the likelihood of compromise was found to be 19%, then the rating system may ascribe a “medium” confidentiality risk rating to the infrastructure component.

(134) Further, if the CIA classification includes a confidentiality value of 4 for an infrastructure component (e.g., class D), and the likelihood of compromise was found to be 65%, then the rating system may ascribe a “high” confidentiality risk rating to the infrastructure component.

(135) Ratings for integrity and availability may be obtained in similar manners. These ratings may be used to select the extent of remediations to be performed as well as convey the CIA risks presented by the infrastructure component.

(136) Turning to FIG. 8B, a diagram of dashboard **840** in accordance with an embodiment is shown. Dashboard **840** may be displayed to a person to inform the person of risk and allow user feedback to be obtained.

(137) When displayed to the person, dashboard **840** may be populated with graphical elements to convey information regarding CIA risk presented by an infrastructure component. To convey the information, dashboard **840** may include system identifier **842**, any number of type indicators (e.g., **844**) and any number of rating indicators (e.g., **846**). Each of these portions of dashboard **840** is discussed below.

(138) System identifier **842** may be an identifier of an infrastructure component for which CIA risk information is displayed. The identifier may be a name that conveys to a person for which infrastructure component information is being displayed.

(139) The type indicators (e.g., **844**) may indicate different types of CIA risk presented by the infrastructure component.

(140) The rating indicators **846** may be positioned proximate to the type indicators and may be graphical representations indicating the level of the type of the risk indicated by the corresponding risk indicator. In FIG. 8B, different in fill patterns are used to express different ratings. However, other graphical element systems may be used to convey the level of each type of CIA risk presented by an infrastructure component.

(141) While not shown, dashboard **840** may include user controls that through which the user may select which infrastructure components to display information about, to show information regarding different types of risk, to identify infrastructure components with risk levels exceeding certain thresholds, etc.

(142) Additionally, a user may provide user input via dashboard **840** to, for example, select actions to perform as part of remediation. For example, when a user selects a type indicator, a sub-display indicating different actions that may reduce the type of the risk, as well as likely levels of risk reduction if the actions are performed. The actions may be any of those identified as described with respect to FIG. 5 (e.g., remediation actions **516**). The user may select, double click, and/or use other modalities to select the actions to perform as part of a remediation.

(143) Turning to FIG. 9, a flow diagram illustrating a method for obtaining compliance standard data in accordance with an embodiment is shown. The method shown in FIG. 9 may be an expansion of portions of operations **304-308** shown in FIG. 3. The method may be performed in part by computing infrastructure **100**, infrastructure management system **110**, and/or other components of the system shown in FIG. 1.

(144) At operation **900**, a likelihood estimate for an infrastructure component is obtained using standard compliance data. The likelihood estimate may be obtained by subtracting the compliance

quantification(s) specified by the compliance standard data from 100%, or a different quantification may be obtained.

(145) At operation **902**, CIA ratings for an infrastructure component are obtained using the likelihood estimate, CIA classifications for the infrastructure component, and a rating system. The CIA ratings may be obtained by ingesting the likelihood estimate and the CIA classifications into the rating system. The rating system may output the CIA ratings.

(146) At operation **904**, the CIA ratings are used to determine whether the security state of the infrastructure component has changed. The CIA ratings may be used by presenting, using the confidentiality-integrity-availability rating, a dashboard; obtaining, using the dashboard, user input responsive to the confidentiality-integrity-availability rating; and using the user input to make the determination. For example, the user input may indicate whether the security state of the infrastructure component has changed.

(147) The confidentiality-integrity-availability classifications may include a first classification for confidentiality. The first classification may indicate a numerical value within a range (e.g., 1-5) for confidentiality concern for the infrastructure component. Similarly classifications for integrity and availability may also be included in the CIA classifications.

(148) The method may end following operation **904**.

(149) Any of the components illustrated in FIGS. **1-2D**, **5**, and **7-8B** may be implemented with one or more computing devices. Turning to FIG. **10**, a block diagram illustrating an example of a data processing system (e.g., a computing device) in accordance with an embodiment is shown. For example, system **1000** may represent any of data processing systems described above performing any of the processes or methods described above. System **1000** can include many different components. These components can be implemented as integrated circuits (ICs), portions thereof, discrete electronic devices, or other modules adapted to a circuit board such as a motherboard or add-in card of the computer system, or as components otherwise incorporated within a chassis of the computer system. Note also that system **1000** is intended to show a high level view of many components of the computer system. However, it is to be understood that additional components may be present in certain implementations and furthermore, different arrangement of the components shown may occur in other implementations. System **1000** may represent a desktop, a laptop, a tablet, a server, a mobile phone, a media player, a personal digital assistant (PDA), a personal communicator, a gaming device, a network router or hub, a wireless access point (AP) or repeater, a set-top box, or a combination thereof. Further, while only a single machine or system is illustrated, the term “machine” or “system” shall also be taken to include any collection of machines or systems that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein.

(150) In one embodiment, system **1000** includes processor **1001**, memory **1003**, and devices **1005-1007** via a bus or an interconnect **1010**. Processor **1001** may represent a single processor or multiple processors with a single processor core or multiple processor cores included therein. Processor **1001** may represent one or more general-purpose processors such as a microprocessor, a central processing unit (CPU), or the like. More particularly, processor **1001** may be a complex instruction set computing (CISC) microprocessor, reduced instruction set computing (RISC) microprocessor, very long instruction word (VLIW) microprocessor, or processor implementing other instruction sets, or processors implementing a combination of instruction sets. Processor **1001** may also be one or more special-purpose processors such as an application specific integrated circuit (ASIC), a cellular or baseband processor, a field programmable gate array (FPGA), a digital signal processor (DSP), a network processor, a graphics processor, a network processor, a communications processor, a cryptographic processor, a co-processor, an embedded processor, or any other type of logic capable of processing instructions.

(151) Processor **1001**, which may be a low power multi-core processor socket such as an ultra-low voltage processor, may act as a main processing unit and central hub for communication with the

various components of the system. Such processor can be implemented as a system on chip (SoC). Processor **1001** is configured to execute instructions for performing the operations discussed herein. System **1000** may further include a graphics interface that communicates with optional graphics subsystem **1004**, which may include a display controller, a graphics processor, and/or a display device.

(152) Processor **1001** may communicate with memory **1003**, which in one embodiment can be implemented via multiple memory devices to provide for a given amount of system memory. Memory **1003** may include one or more volatile storage (or memory) devices such as random access memory (RAM), dynamic RAM (DRAM), synchronous DRAM (SDRAM), static RAM (SRAM), or other types of storage devices. Memory **1003** may store information including sequences of instructions that are executed by processor **1001**, or any other device. For example, executable code and/or data of a variety of operating systems, device drivers, firmware (e.g., input output basic system or BIOS), and/or applications can be loaded in memory **1003** and executed by processor **1001**. An operating system can be any kind of operating systems, such as, for example, Windows® operating system from Microsoft®, Mac OS®/iOS® from Apple, Android® from Google®, Linux®, Unix®, or other real-time or embedded operating systems such as VxWorks.

(153) System **1000** may further include IO devices such as devices (e.g., **1005**, **1006**, **1007**, **1008**) including network interface device(s) **1005**, optional input device(s) **1006**, and other optional IO device(s) **1007**. Network interface device(s) **1005** may include a wireless transceiver and/or a network interface card (NIC). The wireless transceiver may be a WiFi transceiver, an infrared transceiver, a Bluetooth transceiver, a WiMax transceiver, a wireless cellular telephony transceiver, a satellite transceiver (e.g., a global positioning system (GPS) transceiver), or other radio frequency (RF) transceivers, or a combination thereof. The NIC may be an Ethernet card.

(154) Input device(s) **1006** may include a mouse, a touch pad, a touch sensitive screen (which may be integrated with a display device of optional graphics subsystem **1004**), a pointer device such as a stylus, and/or a keyboard (e.g., physical keyboard or a virtual keyboard displayed as part of a touch sensitive screen). For example, input device(s) **1006** may include a touch screen controller coupled to a touch screen. The touch screen and touch screen controller can, for example, detect contact and movement or break thereof using any of a plurality of touch sensitivity technologies, including but not limited to capacitive, resistive, infrared, and surface acoustic wave technologies, as well as other proximity sensor arrays or other elements for determining one or more points of contact with the touch screen.

(155) IO devices **1007** may include an audio device. An audio device may include a speaker and/or a microphone to facilitate voice-enabled functions, such as voice recognition, voice replication, digital recording, and/or telephony functions. Other IO devices **1007** may further include universal serial bus (USB) port(s), parallel port(s), serial port(s), a printer, a network interface, a bus bridge (e.g., a PCI-PCI bridge), sensor(s) (e.g., a motion sensor such as an accelerometer, gyroscope, a magnetometer, a light sensor, compass, a proximity sensor, etc.), or a combination thereof. IO device(s) **1007** may further include an imaging processing subsystem (e.g., a camera), which may include an optical sensor, such as a charged coupled device (CCD) or a complementary metal-oxide semiconductor (CMOS) optical sensor, utilized to facilitate camera functions, such as recording photographs and video clips. Certain sensors may be coupled to interconnect **1010** via a sensor hub (not shown), while other devices such as a keyboard or thermal sensor may be controlled by an embedded controller (not shown), dependent upon the specific configuration or design of system **1000**.

(156) To provide for persistent storage of information such as data, applications, one or more operating systems and so forth, a mass storage (not shown) may also couple to processor **1001**. In various embodiments, to enable a thinner and lighter system design as well as to improve system responsiveness, this mass storage may be implemented via a solid state device (SSD). However, in other embodiments, the mass storage may primarily be implemented using a hard disk drive (HDD)

with a smaller amount of SSD storage to act as a SSD cache to enable non-volatile storage of context state and other such information during power down events so that a fast power up can occur on re-initiation of system activities. Also a flash device may be coupled to processor **1001**, e.g., via a serial peripheral interface (SPI). This flash device may provide for non-volatile storage of system software, including a basic input/output software (BIOS) as well as other firmware of the system.

(157) Storage device **1008** may include computer-readable storage medium **1009** (also known as a machine-readable storage medium or a computer-readable medium) on which is stored one or more sets of instructions or software (e.g., processing module, unit, and/or processing module/unit/logic **1028**) embodying any one or more of the methodologies or functions described herein. Processing module/unit/logic **1028** may represent any of the components described above. Processing module/unit/logic **1028** may also reside, completely or at least partially, within memory **1003** and/or within processor **1001** during execution thereof by system **1000**, memory **1003** and processor **1001** also constituting machine-accessible storage media. Processing module/unit/logic **1028** may further be transmitted or received over a network via network interface device(s) **1005**.

(158) Computer-readable storage medium **1009** may also be used to store some software functionalities described above persistently. While computer-readable storage medium **1009** is shown in an exemplary embodiment to be a single medium, the term “computer-readable storage medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) that store the one or more sets of instructions. The terms “computer-readable storage medium” shall also be taken to include any medium that is capable of storing or encoding a set of instructions for execution by the machine and that cause the machine to perform any one or more of the methodologies of embodiments disclosed herein. The term “computer-readable storage medium” shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media, or any other non-transitory machine-readable medium.

(159) Processing module/unit/logic **1028**, components and other features described herein can be implemented as discrete hardware components or integrated in the functionality of hardware components such as ASICs, FPGAs, DSPs or similar devices. In addition, processing module/unit/logic **1028** can be implemented as firmware or functional circuitry within hardware devices. Further, processing module/unit/logic **1028** can be implemented in any combination hardware devices and software components.

(160) Note that while system **1000** is illustrated with various components of a data processing system, it is not intended to represent any particular architecture or manner of interconnecting the components; as such details are not germane to embodiments disclosed herein. It will also be appreciated that network computers, handheld computers, mobile phones, servers, and/or other data processing systems which have fewer components or perhaps more components may also be used with embodiments disclosed herein.

(161) Some portions of the preceding detailed descriptions have been presented in terms of algorithms and symbolic representations of operations on data bits within a computer memory. These algorithmic descriptions and representations are the ways used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. An algorithm is here, and generally, conceived to be a self-consistent sequence of operations leading to a desired result. The operations are those requiring physical manipulations of physical quantities.

(162) It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities. Unless specifically stated otherwise as apparent from the above discussion, it is appreciated that throughout the description, discussions utilizing terms such as those set forth in the claims below, refer to the action and processes of a computer system, or similar electronic

computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission or display devices.

(163) Embodiments disclosed herein also relate to an apparatus for performing the operations herein. Such a computer program is stored in a non-transitory computer readable medium. A non-transitory machine-readable medium includes any mechanism for storing information in a form readable by a machine (e.g., a computer). For example, a machine-readable (e.g., computer-readable) medium includes a machine (e.g., a computer) readable storage medium (e.g., read only memory (“ROM”), random access memory (“RAM”), magnetic disk storage media, optical storage media, flash memory devices).

(164) The processes or methods depicted in the preceding figures may be performed by processing logic that comprises hardware (e.g. circuitry, dedicated logic, etc.), software (e.g., embodied on a non-transitory computer readable medium), or a combination of both. Although the processes or methods are described above in terms of some sequential operations, it should be appreciated that some of the operations described may be performed in a different order. Moreover, some operations may be performed in parallel rather than sequentially.

(165) Embodiments disclosed herein are not described with reference to any particular programming language. It will be appreciated that a variety of programming languages may be used to implement the teachings of embodiments disclosed herein.

(166) In the foregoing specification, embodiments have been described with reference to specific exemplary embodiments thereof. It will be evident that various modifications may be made thereto without departing from the broader spirit and scope of the embodiments disclosed herein as set forth in the following claims. The specification and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense.

Claims

1. A method for managing computing infrastructure, the method comprising: obtaining a compliance information element for an infrastructure component of the computing infrastructure; dynamically processing the compliance information element to update a cross-standard compliance coverage model to obtain an updated cross-standard compliance coverage model; obtaining, using the updated cross-standard compliance coverage model, standard compliance data for a security standard enforced on the infrastructure; obtaining a likelihood estimate for the infrastructure component using the standard compliance data; obtaining, using the likelihood estimate, confidentiality-integrity-availability classifications for the infrastructure, and a rating system, a confidentiality-integrity-availability rating for the infrastructure component; making a determination, using the confidentiality-integrity-availability rating, regarding whether the infrastructure has undergone a change in compliance with the security standard; and in an instance of the determination where the infrastructure has undergone a change in compliance with the security standard: performing an action set to manage an impact of the change in compliance with the security standard.
2. The method of claim 1, wherein using the confidentiality-integrity-availability rating to make the determination comprises: presenting, using the confidentiality-integrity-availability rating, a dashboard; obtaining, using the dashboard, user input responsive to the confidentiality-integrity-availability rating; and using the user input to make the determination.
3. The method of claim 2, wherein the confidentiality-integrity-availability classifications comprise a first classification for confidentiality, the first classification indicating a numerical value within a range for confidentiality concern for the infrastructure component.
4. The method of claim 3, wherein obtaining the confidentiality-integrity-availability rating

comprises: obtaining a quantification using the numerical value and the likelihood estimate.

5. The method of claim 4, wherein obtaining the confidentiality-integrity-availability rating further comprises: obtaining a second quantification using a second numerical value and the likelihood estimate, the second numerical value being within a range for integrity concern for the infrastructure component.

6. The method of claim 1, wherein the confidentiality-integrity-availability rating comprises three sub-ratings, the sub-ratings being for confidentiality concern, integrity concern, and availability concern for the infrastructure component, respectively.

7. The method of claim 6, wherein each of the three sub-ratings are set by a manager of the infrastructure component.

8. The method of claim 1, wherein the likelihood is a likelihood of the infrastructure component becoming compromised.

9. A non-transitory machine-readable medium having instructions stored therein, which when executed by a processor, cause the processor to perform operations for managing computing infrastructure, the operations comprising: obtaining a compliance information element for an infrastructure component of the computing infrastructure; dynamically processing the compliance information element to update a cross-standard compliance coverage model to obtain an updated cross-standard compliance coverage model; obtaining, using the updated cross-standard compliance coverage model, standard compliance data for a security standard enforced on the infrastructure; obtaining a likelihood estimate for the infrastructure component using the standard compliance data; obtaining, using the likelihood estimate, confidentiality-integrity-availability classifications for the infrastructure, and a rating system, a confidentiality-integrity-availability rating for the infrastructure component; making a determination, using the confidentiality-integrity-availability rating, regarding whether the infrastructure has undergone a change in compliance with the security standard; and in an instance of the determination where the infrastructure has undergone a change in compliance with the security standard: performing an action set to manage an impact of the change in compliance with the security standard.

10. The non-transitory machine-readable medium of claim 9, wherein using the confidentiality-integrity-availability rating to make the determination comprises: presenting, using the confidentiality-integrity-availability rating, a dashboard; obtaining, using the dashboard, user input responsive to the confidentiality-integrity-availability rating; and using the user input to make the determination.

11. The non-transitory machine-readable medium of claim 10, wherein the confidentiality-integrity-availability classifications comprise a first classification for confidentiality, the first classification indicating a numerical value within a range for confidentiality concern for the infrastructure component.

12. The non-transitory machine-readable medium of claim 11, wherein obtaining the confidentiality-integrity-availability rating comprises: obtaining a quantification using the numerical value and the likelihood estimate.

13. The non-transitory machine-readable medium of claim 12, wherein obtaining the confidentiality-integrity-availability rating further comprises: obtaining a second quantification using a second numerical value and the likelihood estimate, the second numerical value being within a range for integrity concern for the infrastructure component.

14. The non-transitory machine-readable medium of claim 9, wherein the confidentiality-integrity-availability rating comprises three sub-ratings, the sub-ratings being for confidentiality concern, integrity concern, and availability concern for the infrastructure component, respectively.

15. The non-transitory machine-readable medium of claim 14, wherein each of the three sub-ratings are set by a manager of the infrastructure component.

16. The non-transitory machine-readable medium of claim 9, wherein the likelihood is a likelihood of the infrastructure component becoming compromised.

17. A data processing system, comprising: a processor; and a memory coupled to the processor to store instructions, which when executed by the processor, cause the processor to perform operations for managing computing infrastructure, the operations comprising: obtaining a compliance information element for an infrastructure component of the computing infrastructure; dynamically processing the compliance information element to update a cross-standard compliance coverage model to obtain an updated cross-standard compliance coverage model; obtaining, using the updated cross-standard compliance coverage model, standard compliance data for a security standard enforced on the infrastructure; obtaining a likelihood estimate for the infrastructure component using the standard compliance data; obtaining, using the likelihood estimate, confidentiality-integrity-availability classifications for the infrastructure, and a rating system, a confidentiality-integrity-availability rating for the infrastructure component; making a determination, using the confidentiality-integrity-availability rating, regarding whether the infrastructure has undergone a change in compliance with the security standard; and in an instance of the determination where the infrastructure has undergone a change in compliance with the security standard: performing an action set to manage an impact of the change in compliance with the security standard.

18. The data processing system of claim 17, wherein using the confidentiality-integrity-availability rating to make the determination comprises: presenting, using the confidentiality-integrity-availability rating, a dashboard; obtaining, using the dashboard, user input responsive to the confidentiality-integrity-availability rating; and using the user input to make the determination.

19. The data processing system of claim 18, wherein the confidentiality-integrity-availability classifications comprise a first classification for confidentiality, the first classification indicating a numerical value within a range for confidentiality concern for the infrastructure component.

20. The data processing system of claim 17, wherein the likelihood is a likelihood of the infrastructure component becoming compromised.
